

MATH 7720, HW2, DUE NOV 1

PROBLEM 1, 5PTS

Prove that for $M \in \mathcal{O}_0$ (over $\mathbb{K}$), the following two conditions are equivalent:

- (1) M is filtered by Vermas,
- (2) M is a Λ -graded free $U(\mathfrak{n}^-)$ -module.

PROBLEM 2, 5PTS

Let λ^- denote the anti-dominant element in $\Xi \in \Lambda/(W, \cdot)$. Prove (without using the duality, dual Vermas, etc., only using what we covered in class) that any projective object in $\mathcal{O}^\Xi$ that is also injective is the direct sum of several copies of $P(\lambda^-)$.